ACADEMIC PROJECT PRESENTATION

News & Social Media Analytics Engine

A DSA-based platform for fast search, duplicate detection, story clustering, query correction and scalable content analysis.

GROUP MEMBERS

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Data Structures and Algorithms – 3 (25CS2103E)

A.Y. 2026–2027 | Section 12

Why news & social content needs intelligent analytics

Large-scale digital content creates a search, redundancy and organization challenge.

01 Information Overload

Huge volumes of articles and social posts make manual discovery and review inefficient.

02 Duplicate Content

The same event may appear across multiple sources with identical or slightly changed wording.

03 Related Stories

Different reports can describe one event, requiring efficient similarity and clustering methods.

04 Search & Retrieval

Users need fast pattern matching and relevant retrieval rather than scanning large text collections.

05 Scalability

The engine must remain efficient when workloads grow and support large-scale processing.

Objectives & proposed capabilities

01 Fast Pattern Search

Use KMP and Rabin-Karp to locate patterns efficiently in large text collections.

02 Duplicate Detection

Use suffix arrays and LCP to identify similar document content.

03 Story Clustering

Group related news and social posts into meaningful story clusters.

04 Query Correction

Apply Dynamic Programming for fuzzy matching and query correction.

05 Moderator Assignment

Use network-flow techniques to support assignment and resource allocation.

06 Scalable Analytics

Use randomized and parallel approaches to improve large-workload processing.

Technical stack & requirements

A practical software stack supporting data processing, APIs, analytics and visualization.

Language	Python
Frontend	React.js / HTML / CSS / JavaScript
Backend	FastAPI / Flask
Database	PostgreSQL / MongoDB
DSA Core	KMP, Rabin-Karp, Suffix Arrays, LCP, Dynamic Programming
Analytics	NetworkX / scikit-learn
Visualization	Plotly / D3.js / Chart.js
Deployment	Docker

MINIMUM HARDWARE

Intel Core i5 / Ryzen 5 • 8 GB RAM • 256 GB SSD • Internet

RECOMMENDED

Intel Core i7 / Ryzen 7 • 16 GB+ RAM • 512 GB SSD+

Research background & DSA foundation

Pattern Searching

KMP and Rabin-Karp provide efficient techniques for finding occurrences of patterns in text.

Document Similarity

Suffix arrays and LCP support efficient comparison of text and identification of similar content.

Fuzzy Matching

Dynamic Programming can support approximate matching and query correction.

Large-Scale Assignment

Network-flow techniques can model assignment and resource allocation problems such as moderator distribution.

Scalable Processing

Randomized algorithms and parallel processing can improve performance when processing large workloads.

PROJECT FOCUS **Integrate these DSA techniques into one analytics engine rather than treating each task independently.**

End-to-end processing pipeline

1 COLLECT

News articles + social media posts

2 NORMALIZE

Clean and standardize textual content

3 SEARCH

KMP / Rabin-Karp pattern matching

4 COMPARE

Suffix arrays + LCP similarity

5 CORRECT

Dynamic Programming fuzzy matching

6 CLUSTER

Group related stories and duplicates

7 ASSIGN

Network-flow based moderator allocation

8 ANALYZE

Scalable analytics and dashboard output

INPUT Raw digital content → **DSA ENGINE** → **OUTPUT** Organized, searchable analytics

Core DSA techniques

01 KMP

Fast pattern matching using a prefix/failure function.

02 Rabin-Karp

Hash-based pattern matching for efficient text search.

03 Suffix Array + LCP

Supports document similarity and repeated-content analysis.

04 Dynamic Programming

Enables fuzzy matching and query correction.

05 Network Flow

Supports assignment and resource-allocation decisions.

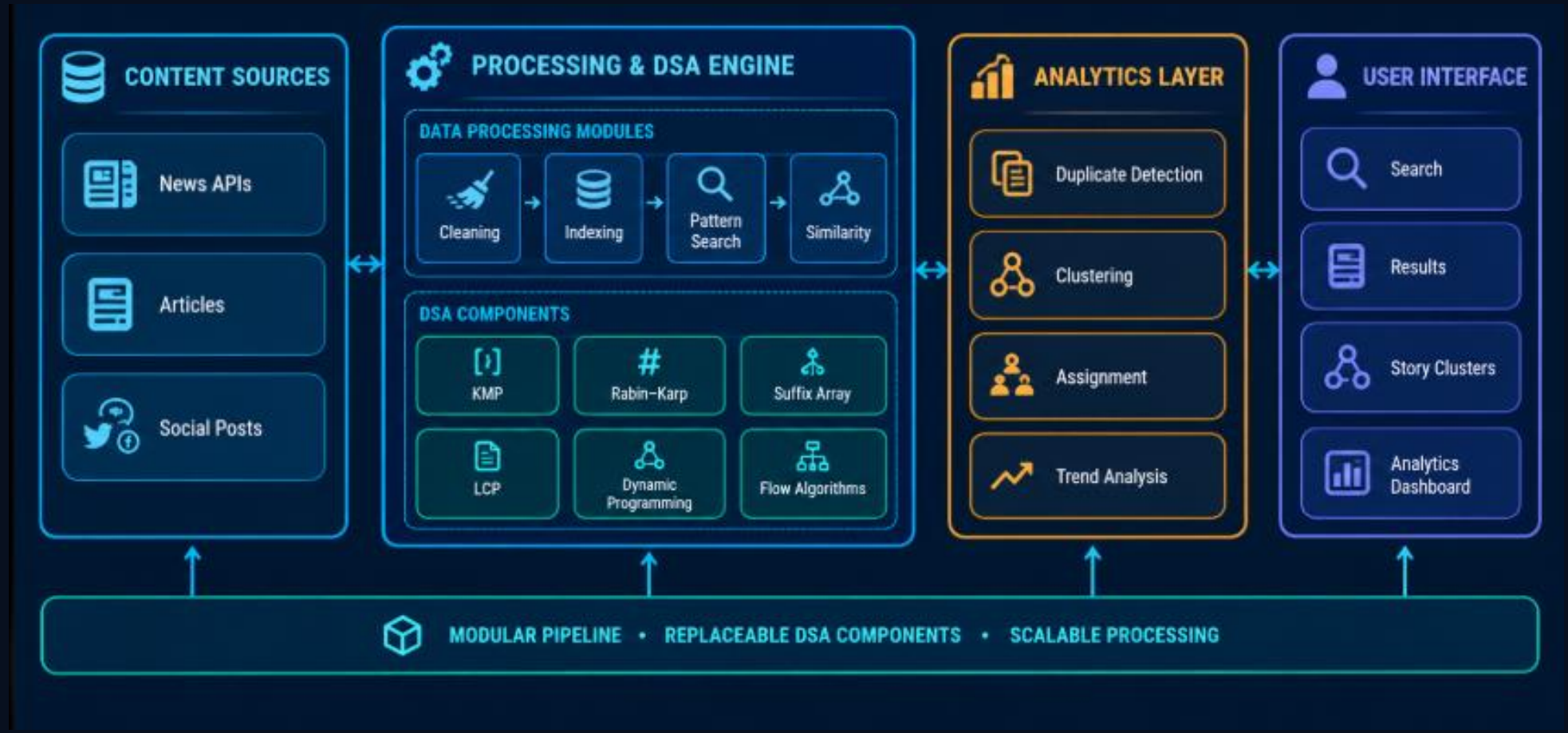
06 Randomized + Parallel

Improves scalability and processing performance for large workloads.

DESIGN PRINCIPLE

Choose the data structure or algorithm according to the operation: search, compare, cluster, assign or scale.

System architecture



A modular pipeline allows individual DSA components to be improved without redesigning the entire system.

Expected outcomes & impact

01 Faster Retrieval

Reduce the effort required to locate relevant content.

02 Lower Redundancy

Identify duplicate information across multiple sources.

03 Better Organization

Cluster related stories and make content easier to explore.

04 Smarter Queries

Correct approximate queries and improve search usability.

05 Efficient Moderation

Support assignment of large content workloads.

06 Scalable Analytics

Process increasing volumes through efficient and parallel approaches.

SEARCH FASTER → DETECT DUPLICATES → CLUSTER STORIES → ANALYZE AT SCALE

Conclusion & future scope

What the project demonstrates

A practical DSA-based engine for large-scale news and social media analysis. It combines fast pattern searching, similarity analysis, fuzzy matching, assignment and scalable processing in one platform.

Why it matters

The approach reduces information redundancy, improves organization and retrieval, and provides a foundation for real-time analytics over rapidly growing digital
content.

Future Scope

Real-time source integration • multilingual analysis • richer story clustering • sentiment/trend analytics • automated summaries • cloud-scale distributed processing

THANK YOU

Questions?